



Antibiotic residues in environment: antimicrobial resistance development, ecological risks, and bioremediation

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Abstract

The overuse of antibiotics and their disposal without processing are leading the environment and its inhabitants towards a serious health emergency. There is abundance of diverse antibiotic resistance genes and bacteria in environment, which demands immediate attention for the effective removal of antibiotics. There are physical and chemical methods for removal, but the generation of toxic byproducts has directed the efforts towards bioremediation for eco-friendly and sustainable elimination of antibiotics from the environment. Various effective and reliable bioremediation approaches have been used, but still antibiotic residues pose a major global threat. Recent developments in molecular and synthetic biology might offer better solution for engineering of microbe–metabolite biodevices and development of novel strains endowed with desirable properties. This review summarizes the impact of antibiotics on environment, mechanisms of resistance development, and different bioremediation approaches.

Keywords Antibiotics · Residues · Resistance · Bioremediation · Sustainable

Introduction

Apart from conventional use in medicine, antibiotics are widely used in genetic engineering, research experiments, animal breeding, and crop production. Globally, the consumption of antibiotics is increasing and is expected to be 200% higher by 2030, than in 2015 when it was 42 billion defined daily doses (DDDs) (Klein et al. 2018). According to the WHO 2018 report, data of 65 countries suggest an overall consumption of antibiotics ranging from 4.4 to 64.4 DDDs per 1000 inhabitants per day. The overuse and misuse of antibiotics have led to widespread appearance of antibiotic resistance genes (ARGs) which in turn have resulted in the development of antibiotic-resistant bacteria (ARB).

These are the emerging environmental pollutants that ultimately hamper the efficacy of lifesaving antibiotic therapies.

The indiscriminate use of antibiotics, especially in the developing countries, has invoked multidrug resistance and virulence in *Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* sp. (ESKAPE) nosocomial pathogens which have become more challenging, mainly in hospital environments (Alemayehu et al. 2019).

Veterinary antibiotic exposure has been linked to liver damage and childhood obesity and other serious health problems identified as a result of ARGs (Park et al. 2020). To address the concerns about enhanced antibiotic concentrations in aquatic and soil environments, various physical (precipitation, membrane filtration, and adsorption), physicochemical (electrochemical treatment, chemical oxidation, ozonation, photodegradation), chemical, and biological degradation methods have been used (Yang et al. 2021). Bioremediation involves the use of activated sludge, microbial strains, and their enzymatic systems to breakdown complex toxic substances. It is an efficient method for antibiotic degradation as it diminishes, detoxifies, mineralizes, and transforms the highly toxic compounds to less toxic ones in an environmental compatibility mode. It is simple, inexpensive, and highly efficient and, more importantly, does not

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cause secondary pollution in environment, whereas physical, chemical, and photolysis degradation methods are potentially expensive, less efficient, and environment unfriendly and also end up adding more pollutants into the environment (Liu et al. 2021).

There are studies reporting antibiotic removal using bacterial, fungal, and algae species. The antibiotics have also been degraded efficiently using microbial enzymes like flavin mononucleotide (FMN) reductases, flavin-dependent monooxygenases, and laccases. In addition to this, some studies have mentioned efficient degradation of antibiotics using activated sludge (Zou et al. 2020; Cecen and Gul 2020). Efforts are on to find even more efficient and advanced ways of bioremediation that help eliminate the limitations associated with traditional bioremediation techniques. Microbial fuel cells are now gaining as these not only degrade antibiotics but also generate electricity simultaneously. Application of synthetic biology for designing synthetic microbial communities and using genome editing is also a new step taken for the degradation of various pollutants including antibiotics (Stein et al. 2018; Leon-Buitimea et al. 2020). The impact of antibiotics on soil and aquatic environment, the mechanisms of antibiotic resistance development, the bioremediation of different classes of antibiotics, novel approaches, and the future prospective are discussed in this review.

Ecological risks of antibiotics in soil environment

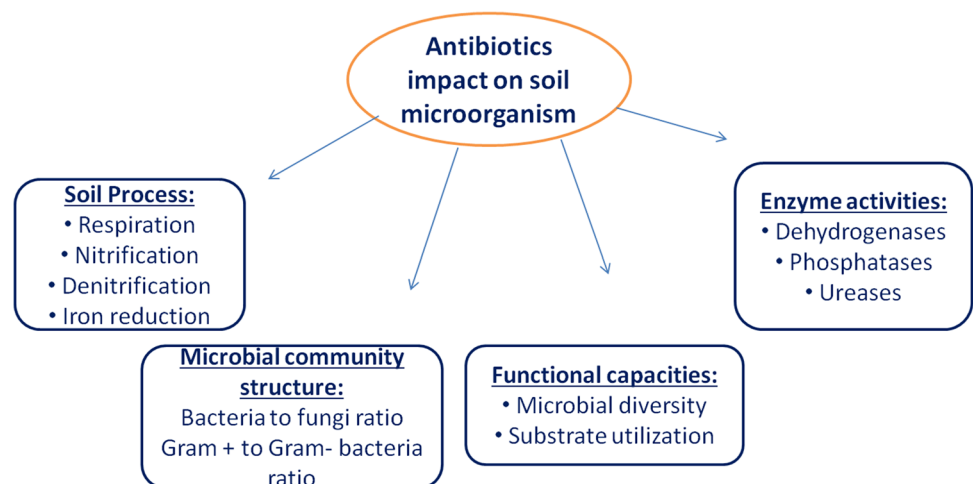
Antibiotics are divided into different classes depending on the mechanism of action and are widely used in treating infections in humans and animals. Although antibiotics are endowed with benefits but as most are not completely absorbed in human and animal bodies, their continuous release and adverse impact on living organisms is a matter

of concern. These antibiotics ultimately reach soil and water through various means. The antibiotics that have the highest concentration in manure are tetracyclines followed by fluoroquinolones and sulfonamides (Masse et al. 2014). These antibiotics affect the abundance of soil microorganisms and their overall microbial and enzyme activity, and the physical and chemical properties of soil like its pH, texture, moisture, concentration of organic matter, and temperature which largely affect the microbial degradation of antibiotics (Fig. 1) (Xu et al. 2016; Ma et al. 2016). In soil, the gene flux and genome reorganization in the microbial community like transfer of ARGs can be induced by even very low concentrations of antibiotics (Grenni et al. 2018).

Cui et al. (2014) reported reduced degradation of ciprofloxacin on increasing its concentration in soil. This may be caused by increased persistence of antibiotics in soil due to inhibition of microbial activity. The persistence of different antibiotics varies with soil composition and doses of antibiotics, and this can be efficiently estimated using a parameter sorption coefficient. It has been found that antibiotics with sorption coefficient values > 4000 l/kg are very persistent while those with < 15 l/kg have low persistence in soil (Crane et al. 2009).

Various studies have reported the impact of antibiotics on individual as well as entire microbial communities and on the soil processes regulated by these microbial populations. Some studies have presented these effects by measuring the parameters like microbial biomass, respiration, and organic matter turnover as these can be correlated with abundance and activity of microbial communities (Cui et al. 2014; Liu et al. 2015; Ma et al. 2016). Some have taken into account the soil processes like methanogenesis, iron reduction, denitrification, and nitrification for assessing the effect of antibiotics on soil microorganisms (Liu et al. 2015; Ma et al. 2016). There are reports of reduced rate of nitrification due to application of a high concentration of oxytetracycline and sulfadiazine in soil (Ma et al. 2016). Cui

Fig. 1 Impact of antibiotics on soil microorganisms



et al. (2014) reported stimulated nitrification in soil microcosm after applying norfloxacin and ciprofloxacin. Soils containing oxytetracycline and sulfamethoxazole residues at concentrations more than 10 mg/kg soil exhibited strong inhibition of iron reduction. The presence of antibiotics in soil affects the growth of soil microorganisms that disturbs the activity of enzymes like urease, phosphatases, and dehydrogenases. Table 1 shows the effect of different antibiotics on soil parameters.

Ecological risks of antibiotics in aquatic environment

Antibiotics are considered water pollutants due to their potential negative impact on aquatic ecosystems and the development of ARGs, which lead to the generation of ARB. Antibiotics are being fed to human and animals on large scale, and they end up in aquatic systems by disposal,

metabolic excretions, or manufacturing. Various sources of antibiotics in groundwater are hospital and industrial discharge, livestock farming, and aquaculture.

Due to their potential adverse effect on aquatic ecosystems, ARB and ARGs can be considered as water contaminants. Various sources of antibiotics in groundwater are hospital and industrial discharge, livestock farming, aquaculture runoff, leaky sewerage, and wastewater effluent (Chen et al. 2017; He et al. 2020).

Antibiotics continuously released into groundwater are taken up by healthy individuals, thereby reducing the ability of antibiotics to treat infections. A major issue is with the inefficiency of wastewater treatment plants (WWTPs) to completely remove antibiotic residues. It has been found that due to the formation of intermediate compounds after the treatment process of parent compound, some antibiotics have been detected at higher concentration in the effluent.

The microbial community plays an important role in the wastewater treatment process, but their growth is inhibited

Table 1 Effect of antibiotics on microbial processes, enzyme activities, and physiological profiles of microbial communities in soil

Antibiotic (conc., in mg/kg)	Effect on soil processes/enzyme activities/microbial community	Reference
Tetracyclines		
Tetracycline (100)	Decreased functional diversity, stimulated basal respiration with no inhibitory effect on respiration	Ma et al. (2014)
Chlortetracycline (1–300)	Inhibition of acid phosphatase	Liu et al. (2009)
Oxytetracycline (1–30)	Decreased nitrification, stimulated dehydrogenase, no effect on phosphatase and urease, and negatively affected the soil microbial community metabolism	Ma et al. (2016)
Oxytetracycline (1–100)	Negatively affected the soil respiration and completely inhibited Fe(III) reduction	Molaei et al. (2017)
Oxytetracycline (5–200)	Declined dehydrogenase, increased richness, and diversity at 50 mg/kg	Unger et al. (2013)
Oxytetracycline (1–217)	Decreased substrate utilization, functional diversity, and evenness	Kong et al. (2006)
Oxytetracycline (1–200)	Decreased phosphatase, urease, arylsulfatase, and dehydrogenase	Chen et al. (2013)
Chlortetracycline (10, 100)	Inhibited dehydrogenase, acid phosphatase, and urease activity	Liu et al. (2014a)
	Diversity and richness of microbial community detrimentally affected	
Chlortetracycline (1–100)	Inhibited dehydrogenase and urease	Liu et al. (2015)
Doxycycline (10)	Microbial diversity affected positively	Wang et al. (2016)
Sulfonamides		
Sulfamonomethoxine (100)	Stimulated basal respiration and functional diversity negatively affected	Ma et al. (2014)
Sulfadiazine (10, 100)	Basal respiration inhibited by 10 mg/kg and 100 mg/kg on the 1 st and 7 th days and found increased after 28 days at 10 mg/kg	Xu et al. (2016)
	Fluorescein diacetate inhibited; dehydrogenase inhibited within 14 days and increased after 28 days at 10 mg/kg	
Sulfamethazine (20, 100)	Increased CO ₂ -C efflux, increased nitrification	Awad et al. (2016)
Others		
Norfloxacin (fluoroquinolone) (5, 10, 30)	No inhibitory effect on soil respiration; nitrogen transformation slightly affected	Yang et al. (2012)
Lincomycin (lincosamide) (5–200)	Low dehydrogenase and higher fluorescein diacetate	Unger et al. (2013)
Penicillin G (β lactam) (10, 600)	Inhibitory effect on catalase, dehydrogenase, phosphatase, and invertase	Akimenko et al. (2015)
Tylosin (macrolide) (100, 600)	Inhibitory effect on catalase, dehydrogenase, phosphatase, and invertase	Akimenko et al. (2015)
Vancomycin (glycopeptide) (1, 10)	Inhibition of dehydrogenase, phosphatase, and urease	Cycon et al. (2018)

by the presence of antibiotics, thereby decreasing the efficiency of WWTPs. Not only groundwater contamination but also the irrigation of agricultural soil with antibiotic-contaminated water can lead to loss of important bacterial groups which are necessary for nitrogen fixation, which in turn affects the crop growth and production (Roose-Amsaleg and Laverman 2015). Yan et al. (2013) performed a series of studies with 0.24 µg/l, 2.1 µg/l, 11 µg/l, or 57,500 µg/l sulfamethoxazole and found a significant decrease in denitrification rate at the 57,500 µg/l dose. Inhibition of denitrification rate was also observed when the concentration of sulfamethazine was increased from 0.05 to 100 µg/l (Hou et al. 2015). Denitrification was found to be more affected than nitrification at the therapeutic antibiotic doses (Marques et al. 2021; deVries and Zhang 2016).

The aquatic conditions significantly affect the biodegradation of antibiotics. Antibiotics are not completely degraded in groundwater and may convert to hazardous metabolites because of less abundant and less diverse microbial community as compared to soil (Sui et al. 2015; Lapworth et al. 2012). Sulfamethoxazole is the most detected antibiotic in groundwater among all the sulfonamides. Its huge occurrence can be due to its fast mobility, low biodegradability, and low affinity to soil sorption (Boy-Roura et al. 2018). Although quinolones have a high affinity to soil sorption, their degradation is slowed down due to anaerobic conditions encountered when these enter the groundwater (Ma et al. 2015). The lower concentration of macrolides in groundwater might be due to their less consumption or their limited passage into groundwater due to geochemical processes (Carvalho and Santos 2016). The distribution coefficient (K_d) determines the sorption and is critical for controlling the movement of antibiotics to groundwater (Yang et al. 2017a).

The rivers and lakes act as reservoirs for spreading antibiotic resistance to opportunistic pathogens. The presence of antibiotics, viz. tetracycline, ampicillin, chloramphenicol, and penicillin, is indicated by the high concentrations of ARB resistant to these antibiotics in freshwater systems (Tacao et al. 2015). The dams, ponds, and springs exhibit a relatively low concentration of ARB (Nnadozie and Odume 2019). Various studies have reported β -lactam, tetracycline, chloramphenicol, aminoglycoside, sulfonamide, and macrolide–lincosamide–streptogramin resistance genes in freshwater systems (Czekalski et al. 2014, 2015).

Sulfonamide resistance genes (SRGs) are considered significant indicators of freshwater pollution and are extensively focused on. The two most studied SRGs are *sul1* and *sul2*. WWTPs' effluent discharge into freshwater has been associated with *sul1* abundance while inputs from agricultural runoff, religious rituals, open defecation, and discharges are mainly associated with *sul2* abundance (Lu et al. 2015). A study compared and analyzed 206 wastewater

samples for antibiotic residues, ARGs, and ARB in urban (case study A) and rural (case study B) catchment areas in Germany, where sulfonamides, fluoroquinolones, and macrolide were reported the most frequently detected antibiotics; 85.7% (case study A) and 96.4% (case study B) samples were positive for expanded-spectrum beta-lactamase (ESBL) *Escherichia coli*, and 91.4% (case study A) and 88.2% (case study B) samples were positive for ESBL *Klebsiella* spp., *Enterobacter* spp., and *Citrobacter* spp. (KECs) (Voigt et al. 2019).

Urban activities have an impact on marine environment zones. The highest antibiotic concentrations were found in sewage treatment plants (STPs), harbors, and bays, while the lowest antibiotic concentrations were in open seas and high seas due to dilution and less anthropogenic interference. In addition, estuaries had surface water contamination. Controlling antibiotic pollution in water systems is a major concern because these contaminants can persist for long periods of time in underlying aquifers, making remediation difficult (Jurado et al. 2019).

Antibiotic toxicity and aquatic organisms

The toxicity of antibiotics in water systems is affected by a variety of factors, including the length of exposure, concentration, types of species in the aquatic ecosystem, and availability of other contaminants. Short-term antibiotic exposure is often bacteriostatic or bactericidal, but long-term exposure results in the development of resistance genes in microorganisms. Antibiotic contamination of aquatic ecosystems has the greatest impact on microbial communities, particularly algae, which comprise the majority in aquatic ecosystem. These inhibit the replication, transcription/translation, and different metabolic pathways in chloroplasts (Wang et al. 2015).

Plants have the potential to adsorb a wide range of antibiotics from soil and water and pass them on to the food chain, endangering human health. Fish, a part of human food chain, are also impacted by antibiotic contamination depending on their position in food chain, i.e., herbivores have less antibiotics in their bodies than carnivorous fish which have greater antibiotic concentrations owing to biomagnification (Liu et al. 2021). Different impacts of antibiotics on aquatic organisms are shown in Table 2.

Antibiotic biodegradation

This is an environment-friendly approach as it does not lead to secondary pollution. Antibiotic degradation in the environment is primarily accomplished through the degradation/adsorption ability of single or multiple microbial strains, or

Table 2 Effect of antibiotics on aquatic environment and organisms

Antibiotic	Aquatic environment/matrix	Conc.	Effect on aquatic environment	Reference
Fluoroquinolones				
Enrofloxacin, norfloxacin, ciprofloxacin	River water in a lab study, Poland	EC ₅₀ = 0.13 mg/l, EC ₅₀ = 0.22 mg/l, EC ₅₀ = 0.34 mg/l	Toxic effect on <i>Lemna minor</i>	Wagil et al. (2014)
Norfloxacin and sulfamethoxazole	Fresh water in a lab study, China	≥ 0.4 mg/l 0.24 mg/l	Gonadal DNA damage on exposure of norfloxacin and sulfamethoxazole to <i>Carassius auratus</i>	Liu et al. (2014b)
Enrofloxacin	Freshwater/tilapia cage farming in Thailand	1.59 µg/l	Aquatic crustacean <i>Macrobrachium lanchesteri</i> was found to be highly sensitive to enrofloxacin overall	Rico et al. (2014)
Ciprofloxacin	Wastewater/river water in a lab study, Kenya	0.07 µg/l/0.50 µg/l	Green algae found to be sensitive	Ngumba et al. (2016)
Macrolide				
Erythromycin Clarithromycin	Wastewater in a lab scale, Africa	< 0.01 µg/l	Highly toxic effect on algae and cyanobacteria	Moslah et al. (2017)
Sulfonamides				
Sulfamethoxazole	Sea water/wastewater in a treatment plant (WWTP), Psittalia	0.006 µg/l	Toxic effect on algae and cyanobacteria, crustaceans and rotifers	Alygizakis et al. (2016)
Sulfadiazine	Sea water/wastewater in a treatment plant (WWTP), Psittalia	0.002 µg/l	Toxic effect on <i>Daphnia magna</i> , aquatic invertebrates	Alygizakis et al. (2016)
Sulfadimethoxine	River water (effluent) in a field study, China	0.002 µg/l	Highly toxic effect on microalgae and <i>Cladocerans</i>	Dong et al. (2016)
Sulfamethazine	Marine sediment slurry in a lab study, China	0.005–50 µM	Inhibit growth of microbial biomass	Hou et al. (2015)
Others				
Oxytetracycline (tetracyclines)	Freshwater/tilapia cage farming in Thailand	49 µg/l	Affects the growth of invertebrates	Rico et al. (2014)
Nalidixic acid (quinolones)	River water in a lab study, South Africa	3 µg/l	Green algae and plants are found to be sensitive	Gumbi et al. (2017)

through the enzyme systems involved. These two approaches for antibiotic biodegradation are explained in Tables 3 and 4.

Sulfonamides

Sulfonamides are among the most used synthetic antimicrobials in human and veterinary medicine. Hence, their release in environment is quite expected and alarming as it paves the way for selection of resistant bacteria. Various studies have focused on the biodegradation of sulfonamides using different methods of degradation, viz. bacterial and fungal strains and algae, using microbial fuel cells and activated sludge. Three bacterial cultures from pig manure, activated sludge, and sediment samples were isolated and identified within the genera of *Paracoccus*, *Methylobacterium*, and *Kribbella*. The identified strains degraded 50%, 55.2%, and

60% of sulfadiazine, respectively, and 2-aminopyrimidine and 4-hydroxyl-2-aminopyrimidine were identified as main intermediates. These strains were also found to degrade the major intermediate of sulfadiazine and 2-aminopyrimidine. Zhou et al. (2018) investigated the inducing ability of anthraquinone-2,6-disulfonate (AQDS) and riboflavin to enhance the sulfamethoxazole degradation coupled with ferrous reduction using *Shewanella oneidensis* MR-1. Riboflavin showed an inducing effect on sulfamethoxazole degradation, resulting in 35.5% degradation of sulfamethoxazole. A significant role of *Pleurotus eryngii* mycelium was also observed in reducing the concentration of sulfanilamide, sulfapyridine, sulfadiazine, sulfisoxazole, sulfamethoxazole, and sulfadimethoxine by 96.8%, 80.1%, 73.7%, 98.8%, 95.8%, and 86.2%, respectively, in 7 days of incubation (Baran et al. 2021). Enzyme system was also investigated for the bioremediation of antibiotics. Laccase from co-culture

Table 3 Enzymatic biodegradation of different classes of antibiotics

Antibiotic	Microorganism	Enzyme mediator system	Conditions	Removal rate (%)	Reference
Sulfonamides					
Sulfapyridine, sulfathiazole, and sulfamethazine	<i>T. versicolor</i>	Laccase-dimethoxy-phenol	pH 4.5, 25 °C, 72 h	< 94	Rodriguez-Rodriguez et al. (2012)
Sulfadimethoxine, sulfamonomethoxine	<i>Perenniporia</i> strain	Laccase-syringaldehyde	pH 6, 50 °C, 1 h	88–93	Weng et al. (2013)
Sulfadiazine, sulfamethazine, sulfamethoxazole	<i>Echinodontium taxodii</i>	Laccase-2,2'-azino-bis(3-ethylbenzthiazoline-6-sulfonic acid)	pH 5, 30 min	> 95	Shi et al. (2014)
Sulfamethoxazole, sulfathiazole	<i>T. versicolor</i>	Laccase-1-hydroxybenzotriazole	pH 5, 40 °C, 1 h	76–85	Rahmani et al. (2015)
Sulfamethoxazole	<i>Phoma</i> sp. UHH 5-1-03	Laccase-syringaldehyde	pH 5, 22 °C, 22 h	87	Hofmann and Schlosser (2016)
Tetracyclines					
Tetracycline	<i>T. versicolor</i>	Laccase-2,2'-azino-bis(3-ethylbenzthiazoline-6-sulfonic acid)	pH 6, 25 °C, 24 h	56	de Cazes et al. (2015)
Tetracycline	<i>T. versicolor</i>	Laccase, without mediator	pH 7, 20 °C, 18 h	78	Llorca et al. (2015)
Oxytetracycline, tetracycline	<i>Cerrena</i> sp. HYB07	Laccase-2,2'-azino-bis(3-ethylbenzthiazoline-6-sulfonic acid)	pH 6, 25 °C, 12 h	80	Yang et al. (2017b)
Chlortetracycline, doxycycline, oxytetracycline, tetracycline	<i>T. versicolor</i>	Laccase-syringaldehyde	pH 6, 25 °C, 3 h	95–100	Ding et al. (2016)
Fluoroquinolones					
Flumequine	<i>T. versicolor</i>	Laccase-2,2'-azino-bis(3-ethylbenzthiazoline-6-sulfonic acid)	pH 4, 39 °C, 12 h	98	Ashrafi et al. (2016)
Ciprofloxacin	<i>Aspergillus oryzae</i>	Laccase	pH 6, 60 °C, 5 h	51	Sutar and Rathod (2015)
Ofloxacin, ciprofloxacin, norfloxacin	<i>T. versicolor</i>	Laccase-syringaldehyde	pH 6, 25 °C, 3 h	96–100	Ding et al. (2016)
Nitroimidazole					
Metronidazole	<i>T. versicolor</i>	Laccase-syringaldehyde	pH 6, 25 °C, 24 h	25.9	Becker et al. (2016)
Trimethoprim	<i>T. versicolor</i>	Laccase-2,2'-azino-bis(3-ethylbenzthiazoline-6-sulfonic acid)	pH 7, 20 °C, 12 h	60	Kumar and Cabana (2016)
Macrolide					
Erythromycin	<i>T. versicolor</i>	EreB esterase	pH 7, 37 °C, 16 h	52	Llorca et al. (2015)

of *Alcaligenes faecalis* and *Pycnoporus sanguineus* was reported for 100% sulfamethoxazole elimination within 6 h (Li et al. 2016). A cluster of genes encoding FMN reductase and 2-monooxygenases enabled *Microbacterium* sp. BR1 and *Actinomyces* to inactivate sulfonamides by utilizing their aniline moiety for growth. Kim et al. (2019) identified flavin-dependent monooxygenase, catalyzing the degradation of sulfonamide in non-pathogenic environmental

Actinobacteria. *Paenarthrobacter* sp. A01, isolated from sulfamethoxazole-enriched activated sludge reactors, transformed 90.7% of the drug yielding amino-4,6-dimethylpyrimidine as the major biodegradation product. The study also identified *sad* genes that encode for enzymes responsible for degradation of sulfamethoxazole (Cao et al. 2019). Mayans et al. (2021) examined the degradation potential of *Pleurotus ostreatus* in liquid media, fortified with different

Table 4 Microbial biodegradation of different classes of antibiotics

Antibiotic	Microorganism	Source	Concentration (mg/l)	Removal rate (%)	Reference
Sulfonamides					
Sulfamethoxazole	<i>Ochrobactrum</i> sp. SMXPM1-SA1	Wastewater	5	45.2	Zhang et al. (2019)
Sulfamethoxazole	<i>Acinetobacter</i> sp.	Microbial culture collection	30	100	Wang et al. (2018a)
Sulfadimidine	<i>Bacillus cereus</i> J2	Sewage	50	100	Zhang et al. (2019)
Tetracycline	<i>Klebsiella</i> sp. SQY5	Sewage	10–30	67	Shao et al. (2018)
Aureomycin	<i>Achromobacter</i> sp.	Soil	50	65.5	Cheng et al. (2017)
Tetracycline	<i>Raoultella</i> sp. XY-1	Pig farm	20	70.7	Wu et al. (2018)
Tetracycline	<i>Burkholderia cepacia</i>	Pig farm	10–20	82	Hong et al. (2020)
Fluoroquinolones					
Norfloxacin	<i>Staphylococcus caprae</i> NOR-36	Pharmaceutical wastewater	5	92.6	Fu et al. (2017)
Ciprofloxacin	<i>Lactobacillus gasseri</i>	Effluent water from hospital	5	100	Liyanage and Manage (2018)
Ofloxacin	<i>Rhodococcus</i> sp. FR1	Fluorobenzene	0.45	39	Maia et al. (2018)
Ciprofloxacin	<i>Thermus thermophiles</i> C419	Pharmaceutical wastewater	5	57	Pan et al. (2018)
Ciprofloxacin	<i>Bradyrhizobium</i> sp.	Wastewater	0.05	70	Nguyen et al. (2018)
Macrolide					
Tylosin	<i>Achromobacter</i>	Soil	50	96	Zhang et al. (2018)

sulfonamide antibiotics; 91%, 88%, 83%, 85%, and 87% removal of sulfadiazine, sulfathiazole, sulfapyridine, sulfamethazine, and sulfamethoxazole were achieved after 14 days of incubation. Besides, *P. ostreatus* was used to build a biofilter capable of removing 90% of sulfonamide from wastewater in 24 h.

Fluoroquinolones

Fluoroquinolones inhibit the bacterial enzymes topoisomerase IV and gyrase, playing an essential role in DNA replication. Partially metabolized fluoroquinolones leaving the body enter the environment and drive antibiotic resistance development. *Trametes versicolor* degraded ciprofloxacin and norfloxacin with an efficiency of more than 90% after an incubation of 7 days in malt extract liquid medium (Prieto et al. 2011). P₄₅₀ and the laccase enzyme system of *Pycnoporus sanguineus* and *Phanerochaete chrysosporium* co-culture were involved in the bioremoval of a mixture of ciprofloxacin, norfloxacin, and sulfamethoxazole (Gao et al. 2018).

Rusch et al. (2015) explored the microbial transformation of danofloxacin, a synthetic veterinary fluoroquinolone, by the ascomycete *Xylaria longipes*. Danofloxacin *N*-oxide was identified as the metabolite formed within 3 days in fungal-submerged culture. Eighty percent of the initial antibacterial activity was reduced due to the oxidation of terminal nitrogen of the substituted piperazine moiety of the antibiotic. Glutamine synthetase (GS) transformed norfloxacin to *N*-acetylnorfloxacin and three other metabolites, viz.

8-hydroxynorfloxacin, 6-defluoro-6-hydroxynorfloxacin, and desethylen norfloxacin using the *Microbacterium* sp. 4N2-2 isolated from wastewater treatment plant (Kim et al. 2013). The biosurfactant-producing co-culture of *Bacillus clausii* T and *Bacillus amyloliquefaciens* HM618 reported a bio-transformation efficiency of 76.6% and 88.9% for oxytetracycline and chlortetracycline, respectively (Liu et al. 2020). Yang et al. (2020) examined the biodegradation potential of *Pseudomonas* sp. (SF1), *Pseudomonas* sp. (A12), *Bacillus* sp., and *Clostridium* sp. under aerobic and anaerobic environments of wastewater sludge. Under aerobic conditions, *Pseudomonas* sp. (SF1) and *Pseudomonas* sp. (A12) degraded amoxicillin and chlortetracycline by 88.2% and 82.6%, respectively, while 88.1% (amoxicillin) and 85.2% (chlortetracycline) degradations were achieved using *Bacillus* sp. and *Clostridium* sp.

Cephalosporins

Cephalosporins are semi-synthetic broad-spectrum β -lactams. Algae being eukaryotic organisms are not affected by the toxic effects of cephalosporins as seen on bacterial cell walls. It is therefore advantageous to degrade cephalosporins using microalgae. Guo et al. (2016) investigated cephalosporin 7-aminocephalosporanic acid (7-ACA) removal and biodiesel production using the lipid-rich microalgal strains *Chlamydomonas* sp. Tai-03, *Chlorella* sp. Cha-01, and *Mychonastes* sp. YL-02. It was found that adsorption, hydrolysis, and photolysis reactions were mainly responsible for 7-ACA removal. The removal efficiency was

> 75%, especially with the CHA-01 group. Yu et al. (2017) showed the removal of antibiotic ceftazidime and its parent structure 7-ACA using the green alga *Chlorella pyrenoidosa*. The algal treatment removed 92.7% and 96.07% of the two target compounds, respectively. The main metabolites of ceftazidime were Δ^{-3} ceftazidime and *trans*-ceftazidime. Complete removal of cefuroxime and cefpirome was reported using co-culture of *B. clausii* T and *B. clausii* O/C (Kong et al. 2019). Jiang et al. (2019) examined the removal of cefradine in natural water environment in the presence of *Chlamydomonas reinhardtii*. The removal rate increased to 7.34% in the presence of 10×10^4 cells/ml algal density.

Aminoglycosides

Gentamicin (class aminoglycoside) presents a big challenge for its removal from environment due to its heat stability and resistance to alkaline as well as acidic environment. This makes physical and chemical methods inefficient for its degradation. Various biodegradation strategies have been studied using different bacterial and fungal species. A bacterial consortium named AMQD4 that includes *Providencia vermicola*, *Brevundimonas diminuta*, *Alcaligenes* sp., and *Acinetobacter*, isolated from biosolids, removed gentamicin from unsterilized and sterilized sewage in a lab-scale experiment with a removal efficiency of 56.8% and 47.7%, respectively (Liu et al. 2017). *Aspergillus terreus*, isolated from solid waste and sewage water from gentamicin production factory, has removed approximately 95% of gentamicin by day 7 over a wide pH range. It was found that the fungus first absorbed gentamicin and subsequently degraded it (Liu et al. 2016). *B. diminuta* BZC3 was responsible for maximum degradation among all the isolates in AMQD4, and the main degradation product was 3-acetyl gentamicin (Liu et al. 2017).

Antibiotic resistance

Development of resistance against different antibiotics is a public health threat as it helps bacteria become very successful and smart pathogen. It is an evolutionary process and an emergent property of the microbial community in which horizontal gene transfer (HGT) plays a significant role (Cooper et al. 2017). Some of the major mechanisms by which the interspecies microbial community survives antibiotic exposure are collective resistance, collective tolerance, and exposure protection. Collective resistance refers to interactions within a community that enhances the community members' ability to resist the antibiotic action, whereas collective tolerance refers to interactions within a community that alters cell functioning, such as slowing metabolism, and

thus slows the rate of cell death during transient antibiotic exposure. Interactions within a community that protect vulnerable members during antibiotic treatment by lowering the effective concentration of antibiotic are referred to as exposure protection (Meredith et al. 2015; Bottery et al. 2021).

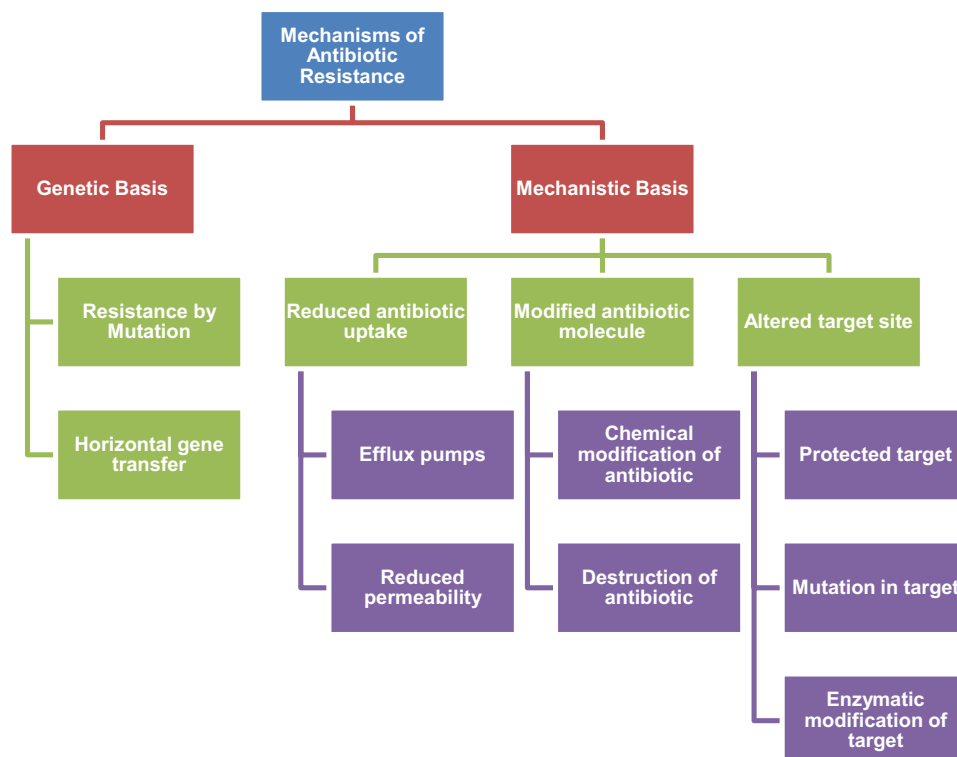
It is accelerated by the antibiotic exposure in environment, agriculture, and healthcare setups (Holmes et al. 2016). A resistance mechanism is induced in bacteria in the presence of antibiotic at sublethal concentration, and the resistance phenotype reverses in the absence of antibiotic to its original state (Krzyzanski and Rao 2017). The basic adaptations in all the microorganisms that develop resistance decreased the permeability of bacterial envelope, biofilm formation, efflux pump modification, destruction or mutation of antibiotic targets in bacteria, and synthesis of enzymes that hydrolyze drugs. The various mechanisms by which microorganisms develop resistance against antibiotics are shown in Fig. 2. *Pseudomonas* spp. and *Acinetobacter* spp. have developed resistance to cephalosporins and carbapenems by decreasing the porin expression (Wu et al. 2016). Resistance to linezolid developed in *Streptococcus pneumoniae* and *S. aureus* due to mutation in one of the multiple copies of the gene encoding 23S rRNA ribosomal subunit target. Methicillin resistance in methicillin-resistant *Staphylococcus aureus* (MRSA) is conferred due to gain of *mecA* gene through the acquisition of staphylococcal cassette chromosome *mec* element (Hiramatsu et al. 2013).

Antibiotic resistance in humans

In India, according to “Scoping report on antimicrobial resistance in India (2017),” *K. pneumoniae*, *A. baumannii*, and most of the strains of *E. coli* and *P. aeruginosa* were reported to be resistant to cephalosporins and fluoroquinolones. As per the Antibiotic Resistance Threats in the United States (CDC 2019) report, around 3 million Americans face antibiotic-resistant infections every year, with 35,000 of them dying. The China Antimicrobial Surveillance Network (CHINET) collected data on bacterial resistance for 13 years starting from the year 2005 and reported an increase in carbapenem-resistant *Acinetobacter* from 31 to 71.4% among which 60% were multidrug resistant alongside a 10-fold (from 2.4 to 24%) increase in carbapenem-resistant *K. pneumoniae* (Qu et al. 2019). The European Antimicrobial Resistance Surveillance Network (EARS-Net 2019), which includes 29 European nations, reported 670,000 infections yearly in member countries due to ARB, with about 33,000 deaths.

Pediatric patients were analyzed in Southern India to explore the antimicrobial resistance patterns of the WHO priority pathogens list (PPL). The most common organism identified was *E. coli* followed by *S. aureus*.

Fig. 2 Mechanisms of the development of antibiotic resistance



Klebsiella spp. (63%) and *E. coli* strains (72%) were resistant to 3rd-generation cephalosporins because of ESBL, and *S. aureus* strains (53%) were resistant to methicillin, whereas the studied specimens tested negative for 3 of the WHO's high priority pathogens, namely *Helicobacter pylori* (clarithromycin resistant), *Neisseria gonorrhoeae* (3rd-generation cephalosporin and fluoroquinolone resistant), and *Campylobacter* spp. (fluoroquinolone resistant) (Mogasale et al. 2021). Kipkorir et al. (2020) collected 184 fecal samples from humans in Kenya, and isolated *E. coli* was highly resistant to tetracycline, ampicillin, amoxicillin, and erythromycin. Among 230 pediatric diarrhea samples, diarrheagenic *E. coli* (DEC) isolates expressed resistance due to ESBL and *AmpC* genes. Among the DEC isolates, multidrug-resistant (MDR) strains were most extensive and exhibited combined resistance to ampicillin, nalidixic acid, tetracycline, and third-generation cephalosporins (Abbasi et al. 2020).

Antibiotic resistance in animals

In a study, nine fattening calf slurry samples and 25 beef cattle farmyard manure (FYM) samples were collected in Belgium and were analyzed for resistant *E. coli* and *Salmonella* strains as well as for 69 antibiotic residues. Seventy percent of the *E. coli* isolates from fattening calf slurry samples were resistant to ampicillin, trimethoprim, tetracycline,

and sulfamethoxazole. From fattening calf slurry samples, 5 *E. coli* isolates were resistant to colistin. Seven fattening calf slurry samples and 7 beef cattle FYM samples were positive for ciprofloxacin-resistant *E. coli*. *Salmonella* was not detected in any of the samples. The six antibiotics, namely ciprofloxacin, enrofloxacin, oxytetracycline, doxycycline, lincomycin, and flumequine, were detected in all flattening calf slurry samples. Antibiotic residues were detected in only 4 of the 25 beef cattle FYM samples. Oxytetracycline was detected in 2 samples along with enrofloxacin, and its metabolite ciprofloxacin was present in different concentrations (Huygens et al. 2021). From poultry meat production chain, 5.3% of *E. coli* isolates were reported to be resistant to various antibiotics such as aminoglycoside, quinolone, tetracycline, trimethoprim, colistin, vancomycin, sulfonamide, and rifampin (Afridi et al. 2020). Amador et al. (2019) examined different poultry samples and found them containing a number of quinolone, tetracycline, chloramphenicol, and sulfonamide resistance genes. Dandachi et al. (2018) carried out a study on MDR gram-negative bacteria isolated from poultry feces in which *E. coli* and *K. pneumoniae* resistant to β -lactams with ESBL and *AmpC* were identified. The spread of resistance to third- and fourth-generation cephalosporins among *Salmonella* spp. can be attributed to the excessive use of ceftiofur and cefquinome in pig farming. One hundred forty-six ARGs have been identified in animal feces from seven large-scale Chinese pig farms in which two farms were under high levels of medication and five under low

levels (Zhao et al. 2018). The spread of colistin resistance is an example of plasmid-mediated transmission, and the first plasmid carrying the colistin resistance gene *mcr-1* was isolated from *Enterobacteriaceae* bacteria from a pig fattening facility (Liu et al. 2016). The feces of beef cattle raised with antibiotics have tested positive for functional *bla_{KPC-2}* (*K. pneumoniae* carbapenemases) (Vikram and Schmidt 2018). The fecal samples collected from dairy cows with mastitis in China tested positive for *K. pneumoniae* containing *bla_{NDM-5}* (He et al. 2017). Tanih et al. (2017) reported the presence of four *Enterococcus* spp.—*Enterococcus durans*, *E. faecium*, *Enterococcus hirae*, and *Enterococcus faecalis*—containing penicillin, erythromycin, tetracycline, and vancomycin resistance genes. An ESBL *K. pneumoniae* has been isolated from cow feces. This strain was MDR, showing resistance to 13 antibiotics (Yang et al. 2019).

Other approaches

Bioremediation has emerged as the most efficient and environment-friendly method of removing persistent pollutants from the environment. But, they are more time consuming, demand more expenditure, and thus provide less output. To obtain better results and eliminate the limitations of bioremediation, alternative technologies are being developed.

Use of synthetic biology for enhancing bioremediation of pollutants has been recently explained (Dvorak et al. 2017; Zhu et al. 2017). Constructing synthetic microbial communities that comprise two or more genetically engineered cell populations can help get rid of limitations linked to natural communities. The prerequisite for constructing synthetic microbial community includes the search for efficient genes from databases (Fajardo et al. 2019). There are various tools that can be used to engineer and control organisms within synthetic microbial communities. The knowledge of quorum sensing and microbial interactions finds great application in bioremediation using synthetic biology. Apart from this, genome editing specifically by CRISPR (clustered regularly interspaced short palindromic repeats) tools has become the priority for many studies.

CRISPR approach

CRISPR-Cas9 can be used to delete, insert, or mutate a gene in a microbial strain of which we wish to increase degradation efficiency (Shapiro et al. 2018; Jaiswal et al. 2019).

CRISPR-Cas system employed along with conventional antibiotics or used as an alternative shows antibacterial effect that may disable ARGs or cleave important domains of genome, thus causing toxicity in bacteria (Jiang et al. 2013). Bikard et al. (2014) reported efficient treatment of *Galleria*

mellonella infection model infected with an *E. coli* strain resistant to chloramphenicol using the CRISPR-Cas system in which Cas9 was directed towards a gene encoding the virulence factor intimin. Intimin is an adhesin which is an important virulence factor in enteropathogenic and enterohemorrhagic *E. coli* strains. Citorik et al. (2014) reported that programmed *gyrA* phagemid exhibited cytotoxicity only against *E. coli* strains with mutated chromosomal *gyrA* that was quinolone resistant and was not cytotoxic for wild-type strains which proves that the CRISPR-Cas system distinguishes between resistant and susceptible strains. Use of Cas9 phagemid has shown successful eradication of MRSA strains from mixed bacterial population (Bikard et al. 2014). The CRISPR-Cas system targets only the virulent bacteria unlike antibiotics that may affect the neighboring bacteria also, and this specificity is an additional advantage of this technology.

Metal nanoparticle

Because of nanosized and distinctive physicochemical properties, nanoparticles have attracted a lot of interest in biotechnology and medical science such as in drug delivery system and production of proteins and monoclonal antibodies (Egbuna et al. 2021). Nanoparticles produced using green approaches like alga-mediated nanoparticles offer an advantage of being less toxic than chemically synthesized nanoparticles. The antibacterial activity of metal nanoparticles against ESKAPE pathogen is well reported (Wang et al. 2017; Lee et al. 2019). Titanium dioxide photocatalytic nanoparticle possesses an antimicrobial effect against multidrug-resistant *E. coli* and methicillin-resistant *S. aureus*. Furthermore, metal nanoparticles have the ability to act synergistically when combined with a wide range of antibiotics currently in use. This combinatorial treatment with antibiotics and nanoparticles is beneficial due to lesser chances of resistance development (Pirrone et al. 2011). Treatment of first-line drug-resistant *Mycobacterium tuberculosis* has been reported by the use of synergism between transition metals and antibiotics (Montelongo-Peralta et al. 2019). A study has found enhanced antibacterial activity of a wide range of antibiotics against gram-negative bacteria and also restored antibiotic susceptibility to a resistant bacterial strain by using silver (Morones-Ramirez et al. 2013). Baaloudj et al. (2021) synthesized and characterized solar light-assisted ZnBi₂O₄ nanoparticles that showed 89% cefixime degradation in 30 min. Other photocatalysis-based ZnO/ α -Fe₂O₃ and nano N-TiO₂/graphene oxide/titan grid sheets were also observed to degrade 99.1% and 80% of cefixime, respectively (Shooshtari and Ghazi 2017; Sheydaei et al. 2018). Electron-hole separation via photon absorption was the

fundamental principle of photocatalytic approaches. The extended electron–hole charge separation improves reaction efficiency, photocatalytic activity, and the mobility of charge carriers to accelerate the chemical response from photoelectrons. Thus, it was inferred that the photocatalytically generated nanoparticles can be used successfully in antibiotic removal from the environment. Despite providing numerous benefits, the toxicity and environmental risk of nanoparticles have received considerable attention in recent years. The interactions of ENPs with other pollutants result in deleterious effect on aquatic and soil organisms such as cell membrane destabilization, decrease monooxygenase activity, and bioaccumulation (Hyseni 2016). Predicted no effect concentrations (PNECs) of nanoparticles was commonly used in order to avoid any environmental risk (Arvidsson 2018). The calculated PNECs for TiO_2 , Ag, CuO, and graphene-based nanomaterials (NMs) in aquatic environments were 1.92×10^{-1} mg/l, 4.85×10^{-5} mg/l, 9.06×10^{-3} mg/l, and 1.15×10^{-2} mg/l, respectively, suggesting the lowest toxicity for TiO_2 NMs, and the highest toxicity for Ag NMs in aquatic environments (Zhao et al. 2020).

Microbial fuel cells

Microbial fuel cells are bioelectrochemical systems that combine microbial and electrochemical processes and have been recently used for degrading antibiotics (Wang et al. 2018a). The use of species like *Geobacter*, *Shewanella*, *Propionibacterium*, *Pseudomonas*, and *Rhodospirillum rubrum* that are electrochemically active is preferred in microbial fuel cells (Rahimnejad et al. 2015). Quejigo et al. (2019) evaluated the mineralization of ^{14}C sulfamethazine in manure with microbial electroremediating cells (MERCs). The native microbial community of manure in the absence of electrodes exhibited 17% of ^{14}C sulfamethazine mineralization after 1 month of incubation. However, in the presence of MERC that provides negative anode potential, the mineralization increased by 10-fold during first 2 weeks. Sediment microbial fuel cell electro-Fenton (SMFC-E-Fenton) system exhibited a 97.4% degradation rate of sulfamethoxazole after a 40-h treatment (Wang et al. 2019).

Sulfamethoxazole and its degradation product 3-amino-5-methylisoxazole (3A5MI) were effectively degraded in microbial fuel cell (MFC) reactors. Within 12 h, approximately 85% of 20 ppm sulfamethoxazole was degraded. Also, the toxic chemical 3A5MI formed during the degradation process could be further mineralized. In addition, biotoxicity of sulfamethoxazole against *Shewanella oneidensis* MR-1 and *E. coli* DH5 α greatly reduced after

MFC treatment and this was shown through an antibacterial activity test (Wang et al. 2018b).

Future perspectives

As most of the studies on bioremediation of antibiotics are limited to research level or lab scale, further research is needed to check the efficacy of these techniques in situ. Microbial communities that are unique to a habitat can influence in situ antibiotic bioremediation, either positively or negatively. As a result, in situ research will provide a realistic view of the potential of specific bioremediation processes when implemented in real-time situations. Since bioremediation approaches are currently limited to the natural functionalities of microbial strains and to their enzymes, improvements are required by integrating synthetic biology approaches with machine learning algorithms. Genetic determinants linked with antibiotic degradation mechanisms must be identified, tailored, or redesigned using genetic engineering tools to advance the enzyme-mediated bioremediation. Inter- and intra-species synthetic/natural microbial communities can also be genetically manipulated to coordinate and improve the cell signaling and quorum sensing for antibiotic degradation. Metabolic engineering has the potential to create tailored and efficient antibiotic degradation pathways in microbes so as to convert antibiotics to non-toxic compounds rather than harmful intermediates.

Conclusion

Antibiotic contamination of ecosystem is going to pose a global threat in the future. Despite having a short half-life, antibiotic persistence in the environment is high due to their lipophilic nature. The use and sale of antibiotics without prescription must be strictly regulated by the competent authorities. The release of antibiotics in the environment poses a serious environmental risk by altering the structure, diversity, and activity of microbial communities. Furthermore, physical adsorption, chemical oxidation, and wastewater treatment plants do not entirely remove antibiotics and are prone to causing the secondary pollution. Bioremediation has been proven as the most effective and indispensable technique for cleaning antibiotic residues. As reported, most of the antibiotics are susceptible to biodegradation using microorganisms and their enzyme systems. Microbial enzymes efficiently transform the toxic antibiotics into less toxic compounds by various mechanisms such as oxidation, reduction, ring-opening, and elimination. In conclusion, utilization of microbes and their enzymes for antibiotic degradation will prove more safety, cost-effectiveness, and

eco-friendliness than other methods for restoring the biological and physiochemical nature of our ecosystems.

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Sanjeev Balda: review and editing

Kirti Kataria: review and editing

Neena Capalash: formal analysis and revision

Prince Sharma: idea, conceptualization, revision, editing, and supervision

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